

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset6_expanded • Generated: 2026-04-14 20:54:43

I. CASE IDENTIFICATION	
Dataset ID	dataset6_expanded
Scan ID	ds-006-exp
Source File	windows_soc_data_6_expanded.csv
Primary Threat	Advanced Persistent Intrusion with Golden Ticket and Multi-Protocol Exfiltration
MITRE Technique	T1195.002
Forensic Confidence	HIGH
Total Log Entries	10,000
Attacked Log Entries	150
Aggregate Risk Score	9,100
Threat Level	CRITICAL

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I EXECUTIVE SUMMARY

- CRITICAL SECURITY INCIDENT OVERVIEW
- A highly sophisticated multi-stage Advanced Persistent Threat (APT) attack was detected involving 7 coordinated attack chains across multiple systems and users.
- Initial access was achieved through a supply chain compromise using a trojanized update binary (MicrosoftUpdateHelper.exe).
- Attackers used PowerShell encoded commands and command shell execution to run malicious payloads and establish control.
- Credential access techniques included LSASS memory dumping (mimikatz) and NTDS.dit extraction from Active Directory.
- A Golden Ticket attack was executed, allowing attackers to forge Kerberos tickets and gain persistent domain-wide access.
- Persistence was maintained using WMI Event Subscriptions, enabling fileless backdoor execution on system events.
- Defense evasion techniques included use of LOLBins (rundll32, regsvr32) and masquerading as legitimate processes like svchost.exe and lsass.exe.
- Extensive domain reconnaissance was performed using net commands and BloodHound for Active Directory enumeration.
- Privilege escalation was achieved via malicious Group Policy modifications and forced propagation using gpupdate.
- Lateral movement was conducted using the EternalBlue (MS17-010) exploit and DoublePulsar backdoor deployment.
- Valid domain accounts were abused post-exploitation to blend malicious activity with legitimate traffic.
- Command and Control (C2) communication used covert tunneling techniques like SSH over HTTPS and WebSocket-based SOCKS5 proxy (Chisel).
- Sensitive data was collected from cloud storage, including AWS S3 bucket enumeration and data extraction.
- Data exfiltration occurred via multiple channels, including OneDrive (Microsoft Graph API) and HTTPS POST requests to attacker-controlled servers.
- The attack demonstrates advanced capabilities combining on-premise Active Directory compromise with cloud exploitation.
- Overall risk level is CRITICAL (91%) with high impact and likelihood of continued attacker presence.
- Immediate response actions include system isolation, credential rotation (especially krbtgt), Kerberos ticket invalidation, patching vulnerabilities, and forensic investigation.
- Failure to respond promptly may lead to full domain compromise, data breaches, and long-term attacker persistence.

II CASE IDENTIFICATION & BACKGROUND

Attack Narrative: Software Supply Chain Compromise and LSASS Credential Dump ; WMI Event Subscription for Persistent Fileless Backdoor ; LOLBin Proxy Execution and Process Masquerading for Defense Evasion ; NTDS.dit Dump and Golden Ticket Kerberos Forgery

Hosts	WIN-SOC-PROD-01, WIN-SOC-PROD-02, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-DC-01, WIN-SOC-PROD-05, WIN-SOC-PROD-06, WIN-SOC-PROD-07, WIN-SOC-PROD-08
Users	NT AUTHORITY\SYSTEM, WIN-SOC\Dev-01, WIN-SOC\User1, WIN-SOC\SvcOps, WIN-SOC\ITAdmin, WIN-SOC\DomainAdmin, WIN-SOC\NetAdmin, WIN-SOC\DBAdmin

III

DESCRIPTION OF EVIDENCE & CHAIN OF CUSTODY

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IV

FORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.
- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V

STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

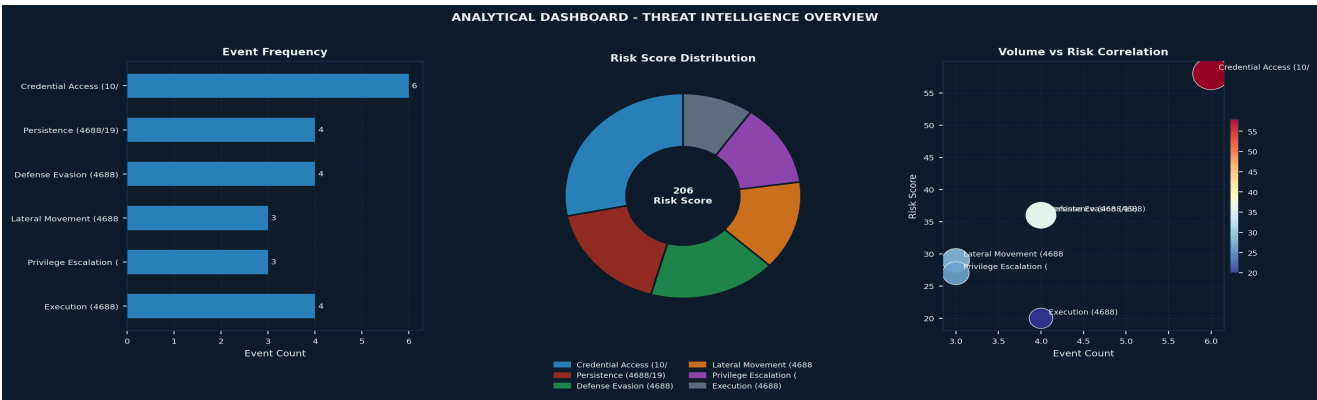


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

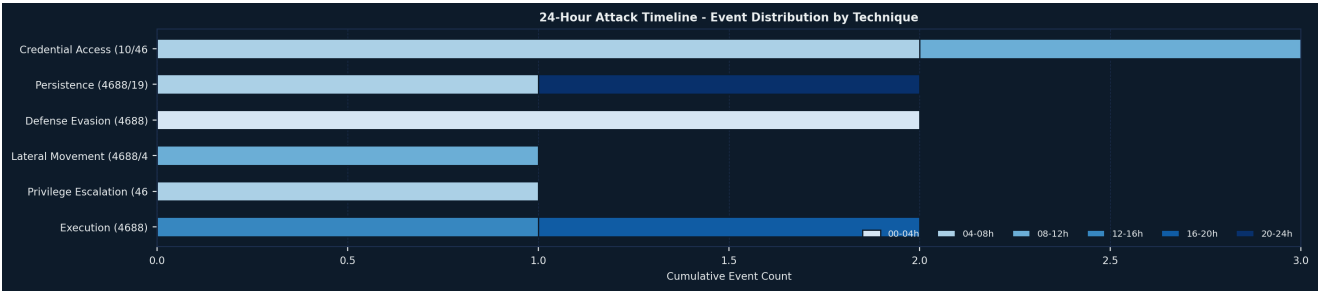


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (10/4662)

LOW

Events: 6	Risk Index: 58	MITRE: T1003.001, T1003.003 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-04-02 00:18:53	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-04-02 00:19:07	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	10
2026-04-02 03:00:00	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	4662
2026-04-02 03:00:50	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	4688

Analysis: Credential Access (10/4662) produced 6 suspicious artifacts with cumulative risk index 58. The mapped technique T1003.001, T1003.003 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4688)

LOW

Events: 4	Risk Index: 36	MITRE: T1218.011, T1036.005 Tactic: Defense Evasion	
Timestamp	Host	User Context	EID
2026-04-02 02:00:00	WIN-SOC-PROD-04	WIN-SOC\ITAdmin	4688
2026-04-02 02:00:45	WIN-SOC-PROD-04	WIN-SOC\ITAdmin	4688
2026-04-02 02:01:30	WIN-SOC-PROD-04	WIN-SOC\ITAdmin	4688
2026-04-02 02:02:15	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688

Analysis: Defense Evasion (4688) produced 4 suspicious artifacts with cumulative risk index 36. The mapped technique T1218.011, T1036.005 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (4688/4624)

LOW

Events: 3	Risk Index: 29	MITRE: T1210 Tactic: Lateral Movement	
Timestamp	Host	User Context	EID
2026-04-02 05:00:00	WIN-SOC-PROD-06	WIN-SOC\DomainAdmin	4688
2026-04-02 05:00:50	WIN-SOC-PROD-07	NT AUTHORITY\SYSTEM	4624
2026-04-02 05:01:40	WIN-SOC-PROD-07	NT AUTHORITY\SYSTEM	4688

Analysis: Lateral Movement (4688/4624) produced 3 suspicious artifacts with cumulative risk index 29. The mapped technique T1210 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX PRIVILEGE ESCALATION - DETAILED ANALYSIS

Privilege Escalation (4688/4741)

LOW

Events: 3	Risk Index: 27	MITRE: T1078.002, T1484.001 Tactic: Privilege Escalation	
Timestamp	Host	User Context	EID
2026-04-02 04:02:00	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	4741
2026-04-02 04:03:00	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	4688
2026-04-02 05:02:30	WIN-SOC-PROD-07	WIN-SOC\DomainAdmin	4688

Analysis: Privilege Escalation (4688/4741) produced 3 suspicious artifacts with cumulative risk index 27. The mapped technique T1078.002, T1484.001 under tactic Privilege Escalation indicates material adversarial behavior requiring priority containment and forensic triage.

X MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (4688/19)

LOW

Events: 4	Risk Index: 36	MITRE: T1546.003 Tactic: Persistence	
Timestamp	Host	User Context	EID
2026-04-02 01:05:00	WIN-SOC-PROD-03	WIN-SOC\SvcOps	4688
2026-04-02 01:05:40	WIN-SOC-PROD-03	WIN-SOC\SvcOps	19
2026-04-02 01:06:20	WIN-SOC-PROD-03	WIN-SOC\SvcOps	20
2026-04-02 01:07:00	WIN-SOC-PROD-03	WIN-SOC\SvcOps	21

Analysis: Persistence (4688/19) produced 4 suspicious artifacts with cumulative risk index 36. The mapped technique T1546.003 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-PROD-01	7	1	LOW
WIN-SOC-DC-01	7	6	MEDIUM

WIN-SOC-PROD-08	6	6	MEDIUM
WIN-SOC-PROD-03	4	4	MEDIUM
WIN-SOC-PROD-04	4	3	LOW
WIN-SOC-PROD-07	3	3	LOW
WIN-SOC-PROD-05	2	0	LOW
WIN-SOC-PROD-02	1	1	LOW
WIN-SOC-PROD-06	1	1	LOW

XII ATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **Software Supply Chain Compromise and LSASS Credential Dump** aligns with primary MITRE mapping **T1195.002** and correlated host-level telemetry.

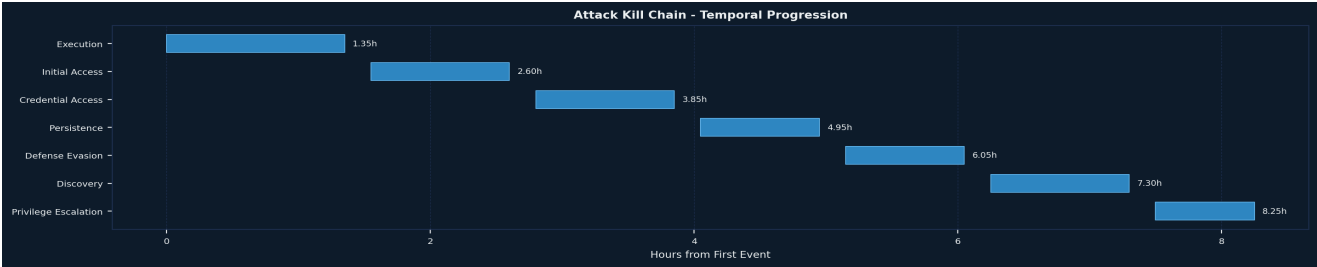
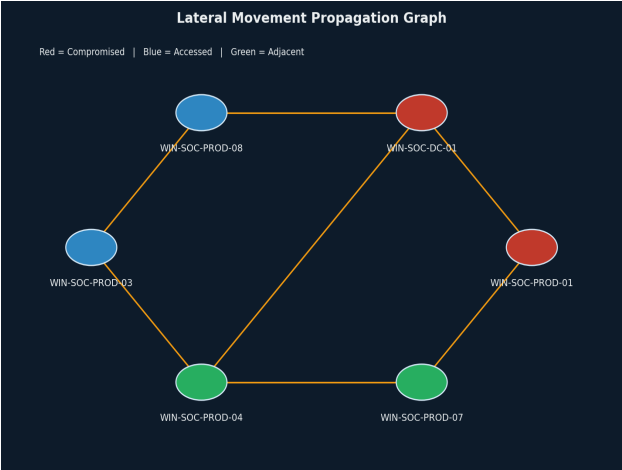


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
chain-ds-006-01	Trojanised Update Binary (MicrosoftUpdateHelper.exe) → PowerShell Encoded IEX Download → cmd.exe Post-Compromise Shell → LSASS Memory Dump (mimikatz.exe)	T1195.002, T1059.001, T1059.003
chain-ds-006-02	WMI ActiveScriptEventConsumer Created → EventFilter Registered (SystemBoot) → CommandLineEventConsumer Bound → FilterToConsumer Binding Activates Persistent Payload on Reboot	T1546.003, T1546.003, T1546.003

chain-ds-0 06-03	Rundll32 LOLBin JavaScript Proxy Execution → Malware Masquerading as svchost.exe from Temp → Regsvr32 Squiblydoo COM Scriptlet → Isass.exe Name Spoofed in Non-Standard Path	T1218.011, T1036.005, T1218.011
chain-ds-0 06-04	ntdsutil AD Database Access → NTDS.dit Extracted via Volume Shadow Copy → Golden Ticket Forged with krbtgt Hash → Golden Ticket Reused Across Multiple Services	T1003.003, T1003.003, T1558.001
chain-ds-0 06-05	net user /domain Enumeration → Domain Admin Group Enumeration → BloodHound AD ACL Discovery → Malicious GPO Startup Script Added → gpupdate Force Push to All Domain Systems	T1087.002, T1087.002, T1484.001
chain-ds-0 06-06	EternalBlue MS17-010 SMB Exploit Launched → SYSTEM Access Gained on PROD-07 → DoublePulsar Backdoor Payload Deployed → Valid Domain Account Abused Post-Exploit	T1210, T1210, T1078.002
chain-ds-0 06-07	SSH over HTTPS C2 Tunnel → Chisel SOCKS5 WebSocket C2 → AWS S3 Bucket Enumeration → S3 Object Download via AWS CLI → Exfiltration to OneDrive via Graph API → HTTPS POST Exfil to C2	T1572, T1530, T1567.002

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security
Process	MicrosoftUpdateHelper.exe	Trojanised Update Binary Executed - Supply Chain Compromise
Process	cmd.exe	Windows Command Shell Spawned by Trojanised Binary
Process	mimikatz.exe	LSASS Dump Post Supply-Chain Compromise
Event ID	10	Sysmon
Process	wmic.exe	WMI Event Subscription Created for Persistence: ActiveScriptEventConsumer
Event ID	19	Sysmon
Event ID	20	Sysmon
Event ID	21	Sysmon
Process	rundll32.exe	Rundll32 LOLBin Proxy Execution: rundll32.exe javascript:..\..\mshtml,RunHTMLApplication
Process	svchost.exe	Masqueraded Malware as svchost.exe Running from C:\Users\ITAdmin\AppData\Local\Temp

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

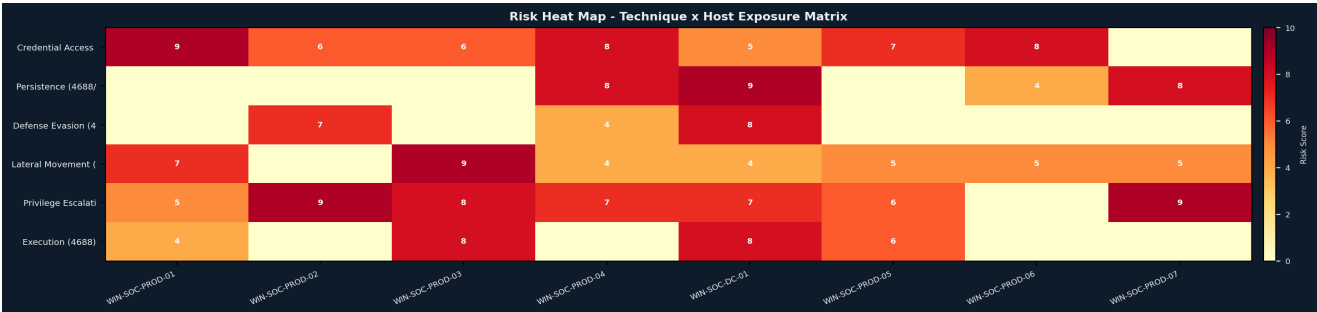


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI

MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

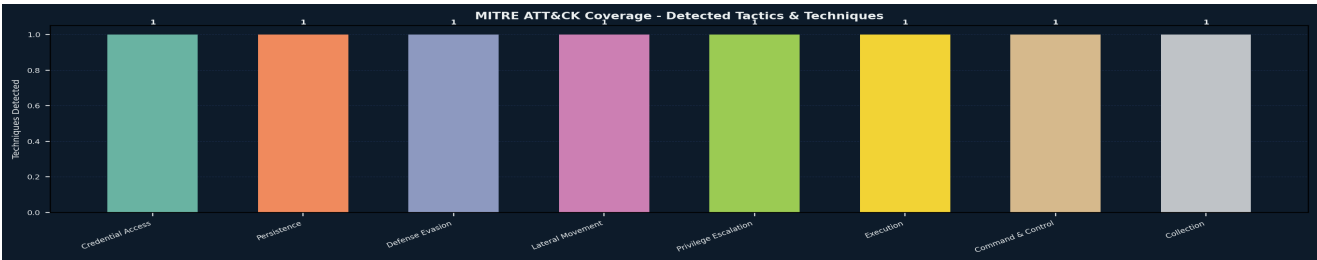


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1003.001, T1003.003	Credential Access (10/4662)	Credential Access
T1546.003	Persistence (4688/19)	Persistence
T1218.011, T1036.005	Defense Evasion (4688)	Defense Evasion
T1210	Lateral Movement (4688/4624)	Lateral Movement
T1078.002, T1484.001	Privilege Escalation (4688/4741)	Privilege Escalation
T1059.001, T1059.003	Execution (4688)	Execution
T1572	Command & Control (5156/3)	Command & Control
T1530	Collection (4688)	Collection

XVII

REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

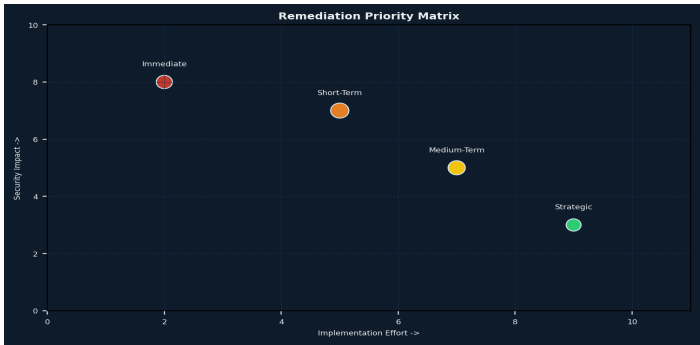


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.
- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII

EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset6_expanded**, the environment exhibits a **CRITICAL** severity posture with aggregate risk index **9,100** across **10,000** records. Dominant indicators include **Credential Access (10/4662) (6 events)**, **Persistence (4688/19) (4 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst

Automated Forensic Engine

Authorizing Officer

Incident Response Division

XIX

ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 20:54:48

XX ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-04-02 00:18:03	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-04-02 00:18:04	WIN-SOC-PROD-01	WIN-SOC\Dev-01	powershell.exe	Encoded IEX Download
2026-04-02 00:18:05	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-04-02 00:18:06	WIN-SOC-PROD-01	WIN-SOC\Dev-01	MicrosoftUpdateHelper.exe	Trojanised Update Binary Executed - Supply Chain Compromise
2026-04-02 00:18:07	WIN-SOC-PROD-01	WIN-SOC\User1	MicrosoftUpdateHelper.exe	Trojanised Binary Spawning cmd.exe - Post-Compromise Execution
2026-04-02 00:18:08	WIN-SOC-PROD-01	WIN-SOC\User1	cmd.exe	Windows Command Shell Spawned by Trojanised Binary
2026-04-02 00:18:53	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump Post Supply-Chain Compromise
2026-04-02 00:19:07	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-04-02 01:05:00	WIN-SOC-PROD-03	WIN-SOC\SvcOps	wmic.exe	WMI Event Subscription Created for Persistence: ActiveScriptEventConsumer
2026-04-02 01:05:40	WIN-SOC-PROD-03	WIN-SOC\SvcOps	wmic.exe	WMI EventFilter Registered: __EventFilter - SystemBoot
2026-04-02 01:06:20	WIN-SOC-PROD-03	WIN-SOC\SvcOps	wmic.exe	WMI EventConsumer Registered: CommandLineEventConsumer - Payload Exec
2026-04-02 01:07:00	WIN-SOC-PROD-03	WIN-SOC\SvcOps	wmic.exe	WMI FilterToConsumer Binding Created - Persistence Active
2026-04-02 02:00:00	WIN-SOC-PROD-04	WIN-SOC\ITAdmin	rundll32.exe	Rundll32 LOLBin Proxy Execution: rundll32.exe javascript:...\mshtml,RunHTMLApplication
2026-04-02 02:00:45	WIN-SOC-PROD-04	WIN-SOC\ITAdmin	svchost.exe	Masqueraded Malware as svchost.exe Running from C:\Users\ITAdmin\AppData\Local\Temp
2026-04-02 02:01:30	WIN-SOC-PROD-04	WIN-SOC\ITAdmin	regsvr32.exe	Regsvr32 COM Scriptlet Execution (Squiblydoo) from Remote URL
2026-04-02 02:02:15	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	lsass.exe	Malicious Binary Masquerading as lsass.exe in Non-Standard Path
2026-04-02 03:00:00	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	ntdsutil.exe	ntdsutil Used to Access NTDS.dit Active Directory Database
2026-04-02 03:00:50	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	ntdsutil.exe	NTDS.dit Volume Shadow Copy Created for Offline Extraction
2026-04-02 03:01:40	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	lsass.exe	Golden Ticket Attack: Forged Kerberos TGT with krbtgt Hash Detected
2026-04-02 03:02:30	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	lsass.exe	Golden Ticket TGT Reuse Across Multiple Services - Anomalous Ticket Lifetime

XXI ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.